



Wireless & unications

- **Motivation**
- **Data transfer**
- **Encapsulation**
- **Security**
- **IPv6**
- **Problems**
- **DHCP**
- **Ad-hoc networks**
- **Routing protocols**





Why Mobile IP?

- What do cellular networks and wireless LANs provide?
 - ❑ Wireless connectivity
 - ❑ Mobility at the data link layer
- What is Dynamic Host Configuration Protocol (DHCP)?
 - ❑ It provides local IP addresses for hosts
- What is Mobile IP?
 - ❑ Transparent connectivity at the network layer
 - ❑ Mobility with local access
- The difference between **mobility** and **nomadicity**!





What is Mobile IP?

- **Mobile IP provides network layer mobility**
- **Provides seamless roaming**
- **“Extends” the home network over the entire Internet**





IP Overview 1/3

➤ IP Addressing :

- ❑ Dotted Decimal Notation: 32 bits (4x8) used to represent IPv4 addresses - 192.19.241.18
- ❑ Network Prefix and Host Portions: p - prefix, h - host, $p + h = 32$. If $p = 24$ then $h = 32 - 24 = 8$. Using above address the network prefix will be 18. For those

255

92.





IP Overview 2/3

➤ IP Routing:

- ❑ Network prefix is used for routing. Routing tables are used to look up next hop and the interface on the router that is to be used.
- ❑ In the routing tables we use the following notation:
target/prefix leng 4, or 192.19.241.192/26.

follow (most specific)
and 7.7.7.0/24 (0 < p < 32 network prefix) and 0.0.0.0/0 (p=0 default) will use entry 2 for an IP packet with destination 7.7.7.1 and entry 3 for destination 192.33.14.12.





IP Overview 3/3

- **Domain Name System (DNS):** used to translate a host name to an IP address. A host sends a query to a server to obtain the IP address of a destination of which it only has the host name.
- **Link Layer Addresses - Address Resolution Protocol (ARP):**
 - ❑ Once a host has destination it then needs

ho
 - ❑ A proxy ARP is a response by a node for another node that cannot respond at the time the request is made (e.g. the node is a mobile node and not on its host net at the time, its home agent will respond in its stead).
 - ❑ A gratuitous ARP, is a reply to no ARP request, used by a node that just joins the network and wants to make its address known. Can be used by a mobile node upon its return to its home net.





Motivation for Mobile IP

➤ IP Routing

- ❑ based on IP destination address, network prefix (e.g. 129.13.42) determines physical subnet
- ❑ change of physical subnet implies change of IP address to have a topologically correct address (standard IP) or needs special entries in

to t

- ❑ does not scale with the number of mobile hosts and frequent changes in the location, security problems

➤ Changing the IP-address?

- ❑ adjust the host IP address depending on the current location
- ❑ almost impossible to find a mobile system, DNS updates take long time
- ❑ TCP connections break, security problems





What Mobile IP does:

- Mobile IP **solves** the following problems:
 - ❑ if a node moves without changing its IP address it will be unable to receive its packets,
 - ❑ if a node changes its IP address it will have to terminate and restart its ongoing connections everytime it moves to a new network area (ne
- Mobilility in the Internet.
- Mobile IP is **not** a complete solution to mobility, changes to the transport protocols need to be made for a better solution (i.e., the transport layers are unaware of the mobile node's point of attachment and it might be useful if, e.g., TCP knew that a wireless link was being used!).





Requirements to Mobile IP (RFC 2002)

➤ Transparency

- ❑ mobile end-systems keep their IP address
- ❑ continuation of communication after interruption of link possible
- ❑ point of connection to the fixed network can be changed

➤ Compatibility

- ❑ mobile end-systems

➤ Security

- ❑ authentication of all registration messages

➤ Efficiency and scalability

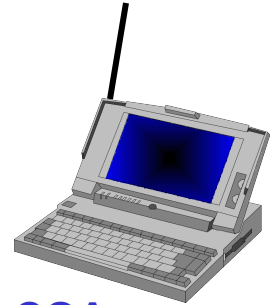
- ❑ only little additional messages to the mobile system required (connection typically via a low bandwidth radio link)
- ❑ world-wide support of a large number of mobile systems in the whole Internet





Mobile IP Terminology

- **Mobile Node (MN)**
 - ❑ system (node) that can change the point of connection to the network without changing its IP address
- **Home Agent (HA)**
 - ❑ system in the home network of the MN, typically a router
 - ❑ registers the location of the MN, tunnels IP datagrams to the COA
- **Foreign Agent (FA)**



ro

- **Care-of Address (COA)**
 - ❑ address of the current tunnel end-point for the MN (at FA or MN)
 - ❑ actual location of the MN from an IP point of view
 - ❑ can be chosen, e.g., via DHCP
- **Correspondent Node (CN)**
 - ❑ communication partner





Mobile IP Operation: Summary

- Consists of 3 steps:
 - ❑ Agent discovery,
 - ❑ Registration, and
 - ❑ Routing/Tunneling





Operation Summary 1/3

- **Agent Advertisement/Discovery:** consists of broadcast messages used by mobiles to detect that they have moved and are required to register with a new FA.
 - ❑ **FAs send agent advertisements**
 - ❑ **MNs can solicit advertisement i**
ve not heard an agent
e other mechanism to
 - ❑ **MN**
A.





Operation Summary 2/3

➤ **Registration:** used by a MN to inform the FA that it is visiting.

- ❑ The new care of address of the MN is sent to the HA.
- ❑ Registration expires, duration is negotiated during registration
- ❑ Mobile must re-r

pas to the FA
which then informs the MN that all is in order and registration
is complete.





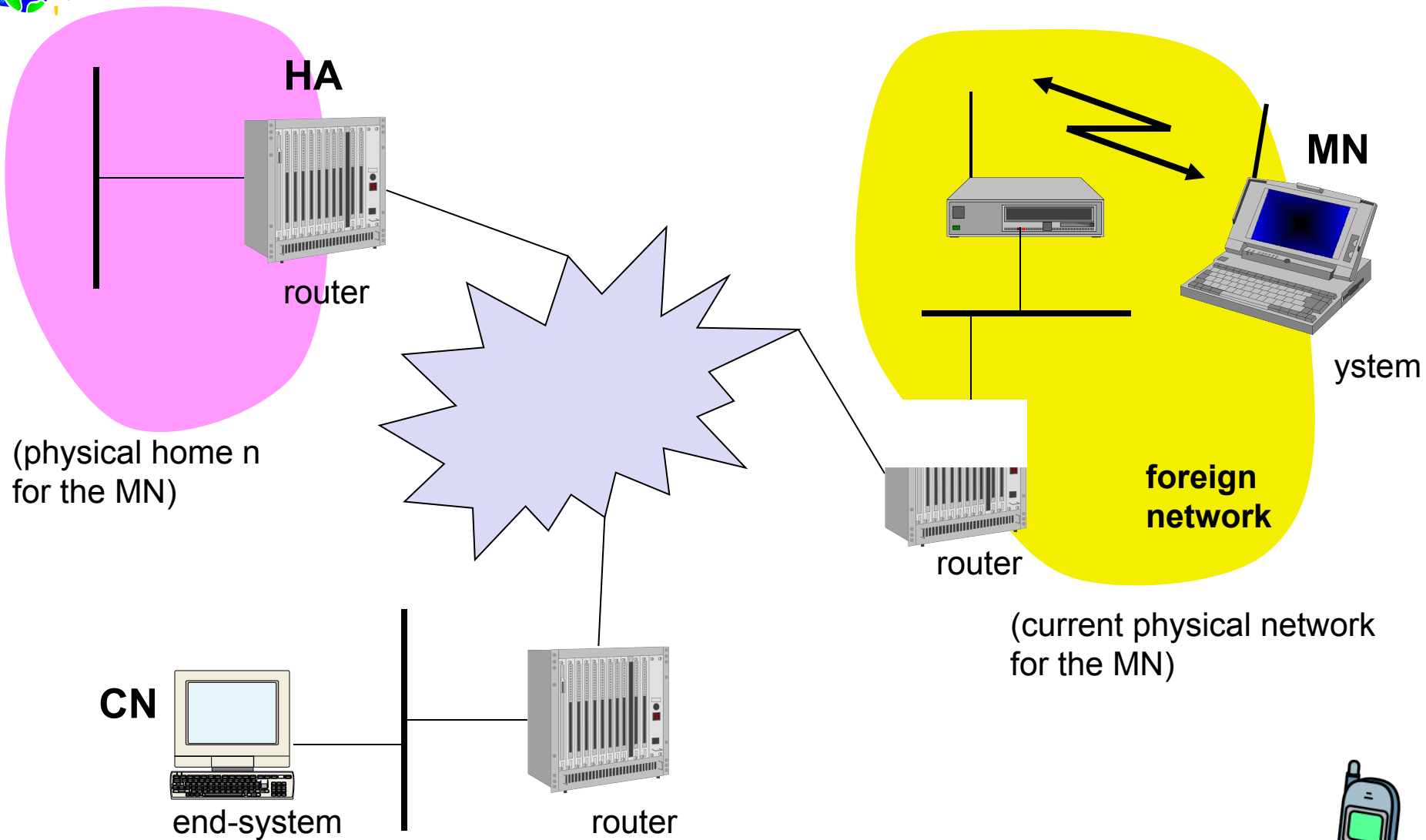
Operation Summary 3/3

- **Routing/Encapsulation/Tunneling:** consists of the delivery of the packets to the mobile node at its current care of address.
 - ❑ **Sender does not need to know that the destination is a MN.**
 - ❑ **HA intercepts all packets destined to the MN and passes them along to the MN using a tunnel.**
 - ❑ **Ref**



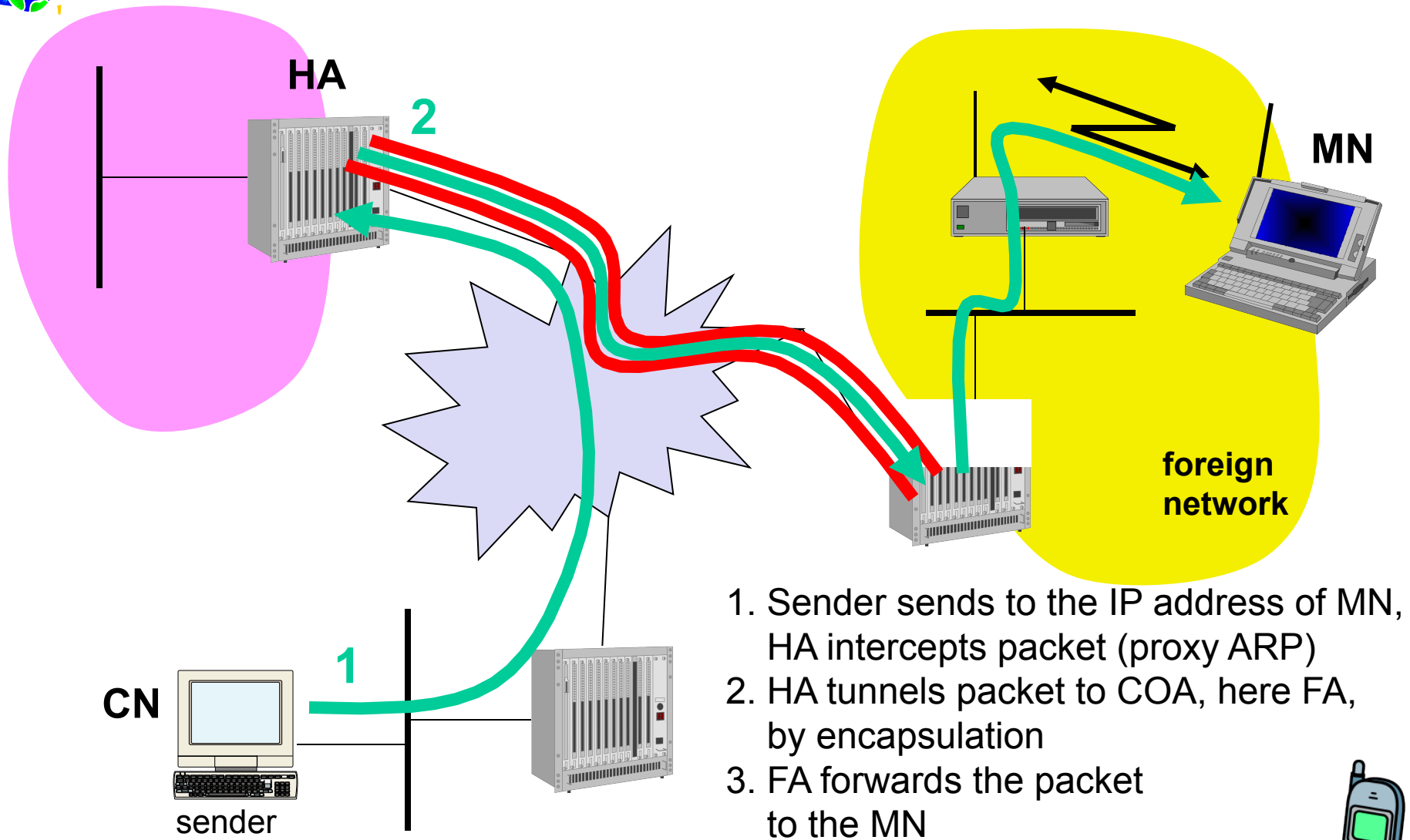


Example network



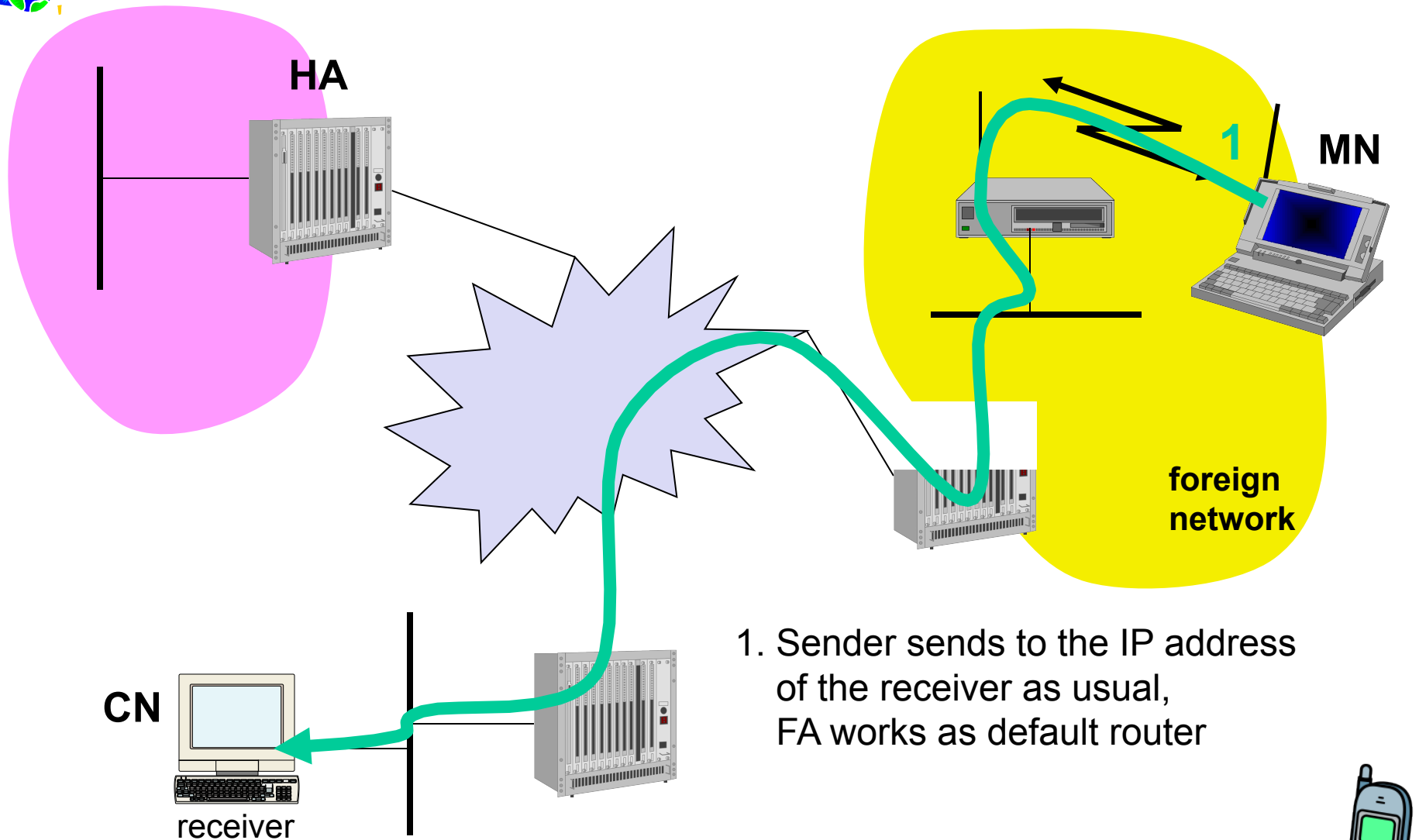


Data transfer to the mobile system



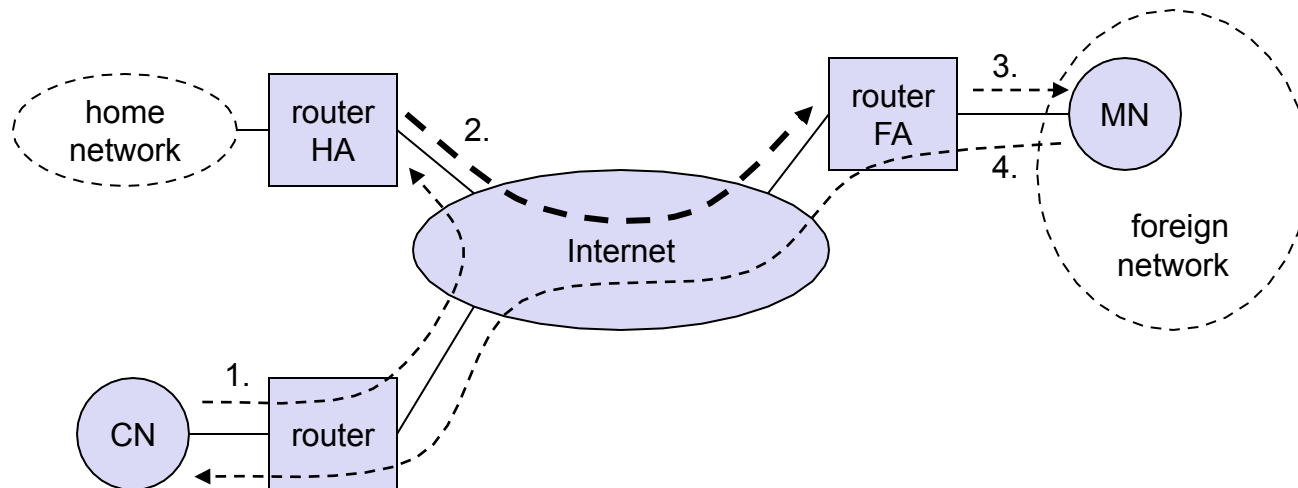
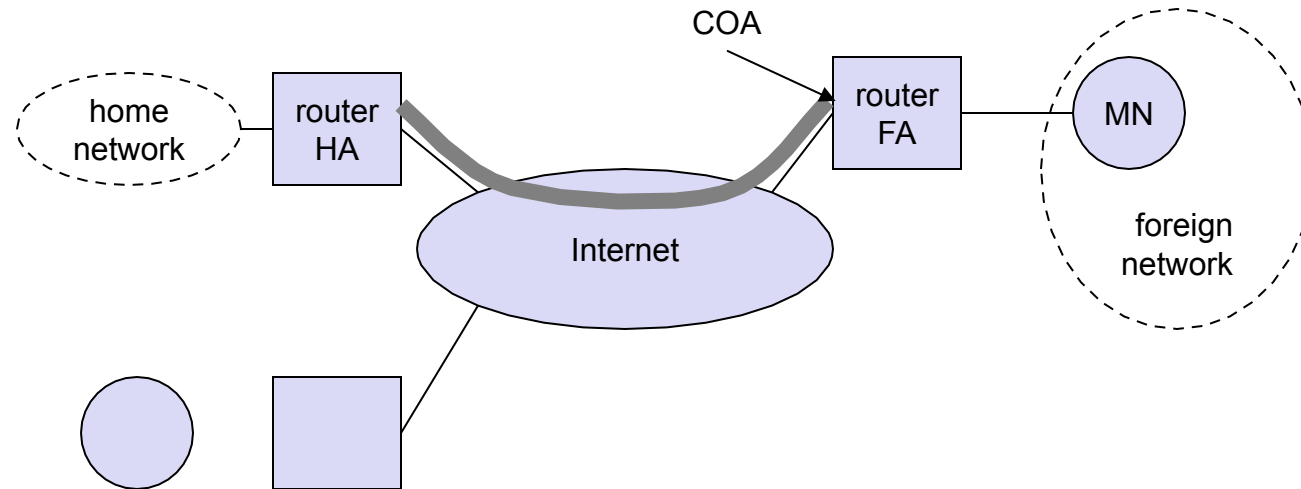


Data transfer from the mobile system





Overview





Network integration

➤ Agent Advertisement Discovery

- ❑ HA and FA periodically send advertisement messages into their physical subnets
- ❑ MN listens to these messages and detects, if it is in the home or a foreign network (standard case for home network)
- ❑ MN reads a COA ent messages

o

MN

- ❑ these actions have to be secured by authentication

➤ Routing/Encapsulation/Tunneling

- ❑ HA advertises the IP address of the MN (as for fixed systems), i.e. standard routing information
- ❑ packets to the MN are sent to the HA,
- ❑ independent of changes in COA/FA





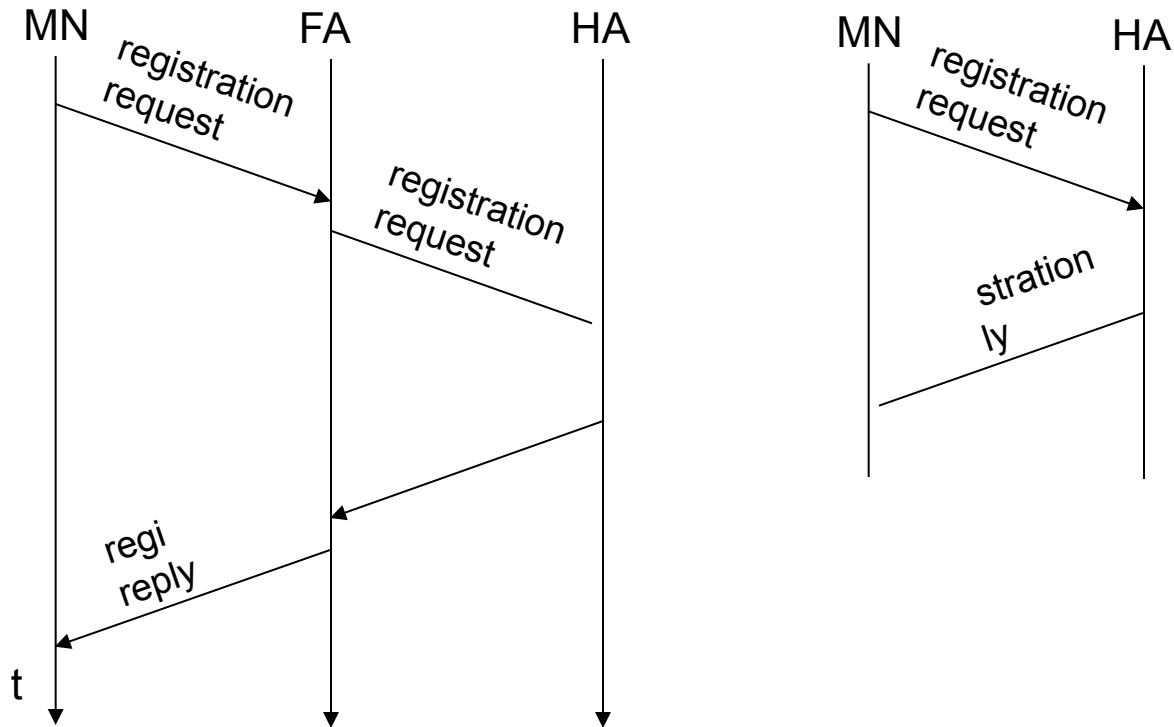
Agent advertisement

0	7	8	15	16	23	24	31
type		code		checksum			
#addresses		addr. size		lifetime			
router address 1							
</							





Registration





Mobile IP registration request

0	7	8	15				16	23	24	31
type		S	B	D	M	G	V	rsv	lifetime	
home address										
home agent										





Processing Registration Messages 1/3

- A MN, depending on which **registration scenario** it is in, will figure what addresses to use in the various fields of the Registration request message.
 - ❑ **Link layer addresses are tricky:**
 - A MN may **not use ARP** if it is using a **FA COA**. It needs to use the address of the **FA** as the **destination** address.
 - If it is using a **coll** to locate the default router using its tion
 - nd it uses its own
 - ❑ **For network layer addresses (i.e., IP addresses):**
 - It uses the **FA address** as **destination** address when using the **FA COA** and its own home address as the source address.
 - If using a **collocated COA** it uses its **COA** as **source** address and the **HA address** as **destination** address. Note that if the 'R' bit is set then it must use the same addresses as for the FA COA scenario.
 - For **de-registration** it uses its own home address as source and the HA address as destination.





Processing Registration Messages 2/3

➤ For the FA:

- ❑ A FA may refuse a Registration request for a number of reasons: lifetime too long, authentication failed, requested tunneling not supported, cannot handle another MN (current load too high).
- ❑ If an FA does not lays it to the HA.
- ❑ So ordered for
use later on: MN link layer address, MN IP address, UDP source port, HA IP address, identification number and requested lifetime.
- ❑ Regarding a Registration reply message, the FA can refuse it and send a decline to the MN if it finds the reply from the HA to be invalid. Otherwise it updates its list of visiting MNs and begins acting on behalf of the MN.





Processing Registration Messages 3/3

➤ For a HA

- ❑ The HA will determine, as the FA did, whether it will accept the request. If it does not it returns a code in the reply message indicating the cause of the failed request.
- ❑ If the request is acepted, the reply is sent back by reversing all the IP address ers.





Routing/Tunneling 1/5

- **Routing a packet to a MN involves the following:**
 - ❑ A router on the home link, possibly the HA, advertises reachability to the network prefix of the MN's home address.
 - ❑ All packets are therefore routed to the MN's home link.
 - ❑ A HA intercepts and tunnels a copy to

(co orwards it
to the MN.





Routing/Tunneling 2/5

- A HA can use one of two methods to intercept a MN's packets:
 - ❑ The HA is a router with multiple network interfaces. In that case it advertises reachability to the MN's home network prefix.
 - ❑ The HA is not a router with multiple network interfaces. It must use ARP

AR the MN's
IP packets. This is to update any ARP caches that hosts and other devices might have.





Routing/Tunneling 3/5

➤ How to ‘**fool**’ the routing table into handling **tunneled packets** at the HA?

- ❑ A **virtual interface** is used to do the encapsulation.
- ❑ A packet destined for the MN is handled by the routing routine as all received IP

t

tha

ulation.

- ❑ Once encapsulation has been performed the packet is sent to be processed by the routing routine **again**. This time the destination address is the COA and it is routed normally.





Routing/Tunneling 4/5

- How to ‘**fool**’ the routing table into handling **tunneled packets** at the FA?
 - ❑ The same procedure is used as above.
 - ❑ A packet coming in with a COA that is one of the FA addresses’ is handled by the r
- le
a
- virt
- ❑ The **virtual interface** consists of a process that decapsulates the packet and re-routes it to the routing routine.
 - ❑ The routing routine routes the packet normally based upon a host specific entry that is the MN’s home address (for which it has the link layer address!).





Routing/Tunneling 5/5

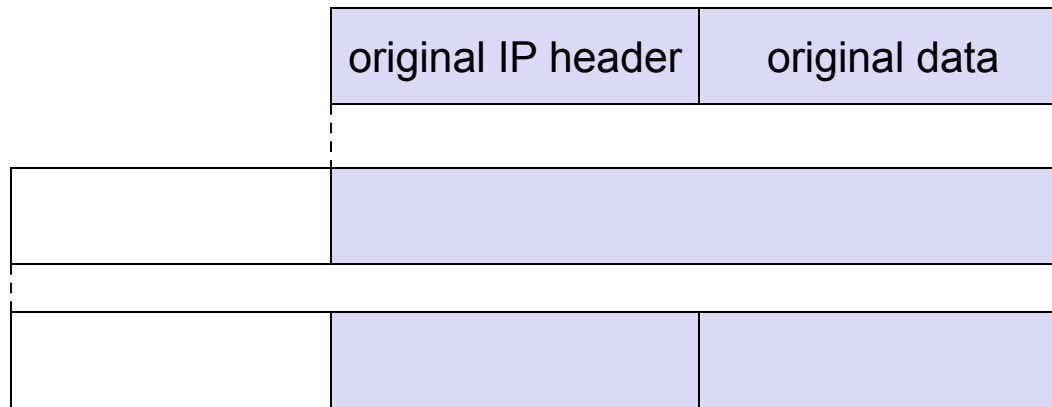
➤ How does a **MN route** its packets?

- ❑ It needs to **find** a router to send all its packets to.
- ❑ It can **select** a router in one of a number of ways dependent upon whether it has a FA COA or a collocated COA.
- ❑ Having a FA COA he MN needs to use it as
t
t
- ❑ If the MN is using a collocated COA it needs to listen for router advertisements or if it hears none, use **DHCP** to find the default router.
- ❑ Determining the **link layer address** is another issue. Collocated COA MNs can use ARP. FA COA must note the link layer address when they receive router advertisements or agent advertisements.





Encapsulation Process





Types of Encapsulation

- Three types of encapsulation protocols are specified for Mobile IP:
 - ❑ **IP-in-IP encapsulation:** required to be supported. Full IP header added to the original IP packet. The new header contains HA address as source and Care of Address as destination.
 - ❑ **Cha** address is maintained as is.
 - ❑ **Generic Routing Encapsulation (GRE):** optional. Allows packets of a different protocol suite to be encapsulated by another protocol suite.
- Type of tunneling/encapsulation supported is indicated in registration.





IP in IP Encapsulation

- IP in IP encapsulation (mandatory in RFC 2003)
 - ❑ tunnel between HA and COA

IP address of HA					
Care-of address COA					
ver.	IHL	TOS		length	
IP identification			flags	fragment offset	
TTL		lay. 4 prot.		IP checksum	
IP address of CN					
IP address of MN					
TCP/UDP/ ... payload					





Minimum Encapsulation

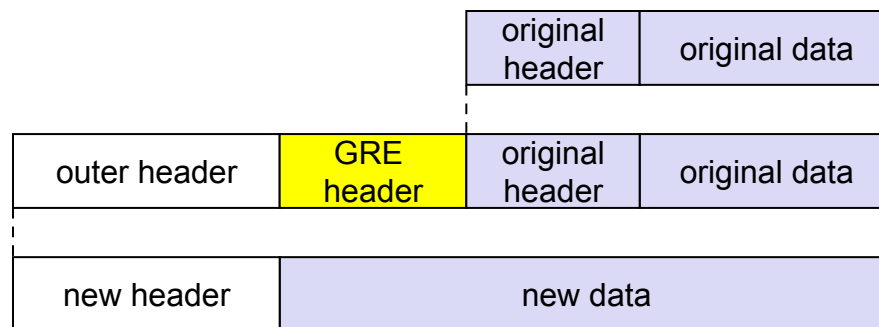
- **Minimal encapsulation (optional)**
 - ❑ avoids repetition of identical fields
 - ❑ e.g. TTL, IHL, version, TOS
 - ❑ only applicable for unfragmented packets, no space left for fragment identification

IP address of HA			
care-of address COA			
lay. 4 protoc.	S	reserved	IP checksum
IP address of MN			
original sender IP address (if S=1)			
TCP/UDP/ ... payload			





Generic Routing Encapsulation



ver.	IHL	TOS	len	
Care				
C	R	K	S	s
rec.	rsv.			
checksum (optional)			offset (optional)	
key (optional)				
sequence number (optional)				
routing (optional)				
ver.	IHL	TOS	length	
IP identification			flags	fragment offset
TTL		lay. 4 prot.	IP checksum	
IP address of CN				
IP address of MN				
TCP/UDP/ ... payload				





Routing techniques

- **Triangle Routing:** tunneling in its simplest form has all packets go to home network (HA) and then sent to MN via a tunnel.
 - ❑ This involves two IP routes that need to be set-up, one original and the second the tunnel route.
 - ❑ Causes unnecessary overhead and adds to the
- **Route** node to learn the current location of the MN and tunnel its own packets directly. Problems arise with
 - ❑ **mobility:** correspondent node has to update/maintain its cache.
 - ❑ **authentication:** HA has to communicate with the correspondent node to do authentication, i.e., security association is with HA not with MN.





Optimization of packet forwarding

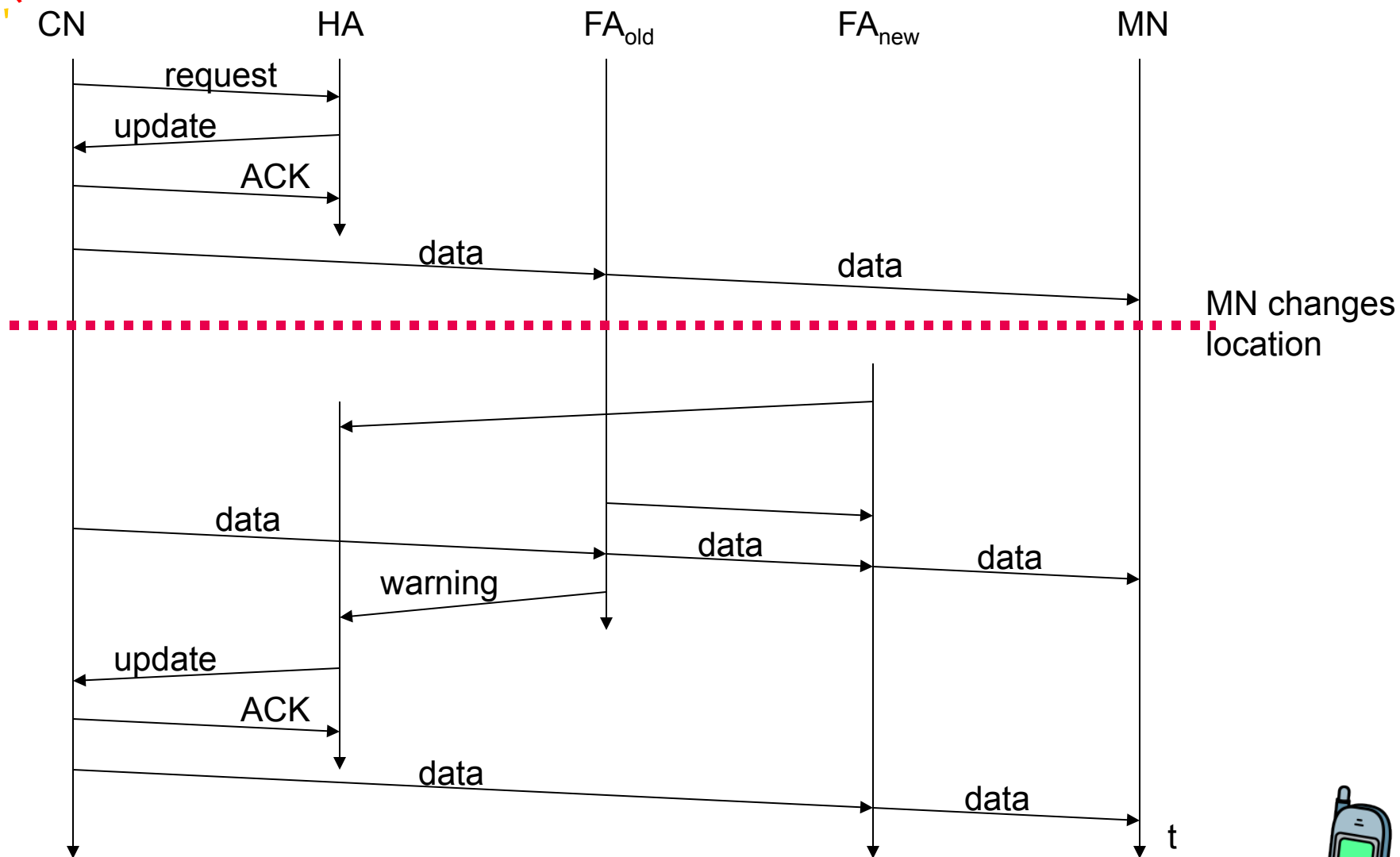
➤ Change of FA

- ❑ packets on-the-fly during the change can be lost
- ❑ new FA informs old FA to avoid packet loss, old FA now forwards remaining packets to new FA
- ❑ this information also enables the old FA to release resources for the MN





Change of foreign agent





Problems with Triangle Routing

- Triangle routing has the MN correspond directly with the CN using its home address as the SA
 - ❑ Firewalls at the foreign network may not allow that
 - ❑ Multicasting: if a MN is to participate in a multicast group, it needs to use a reverse tunnel to maintain its association with the home network

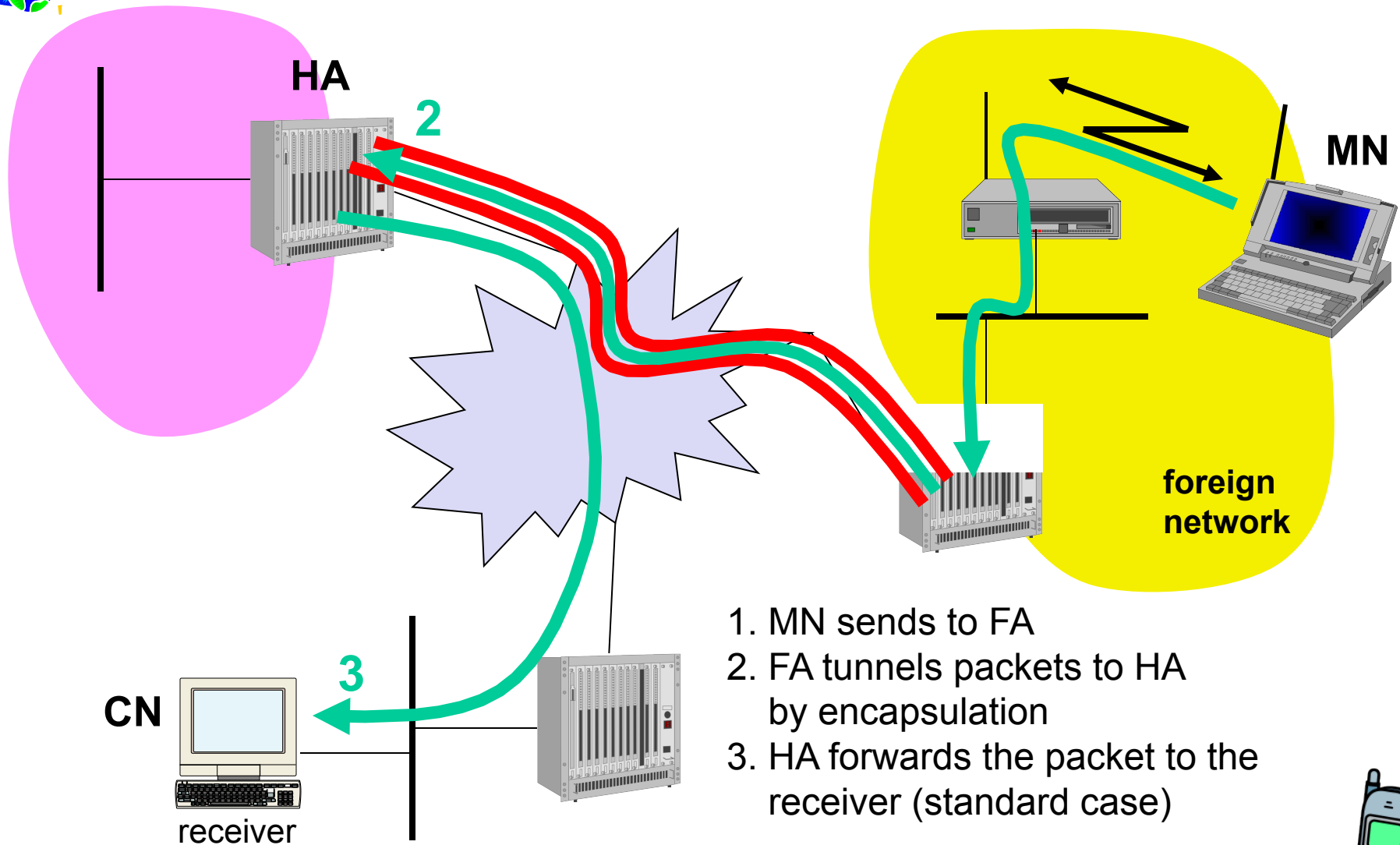
mo a reverse tunnel, it only counts as a single hop. A MN does not want to change the TTL everytime it moves.

- **Solution: reverse tunneling**





Reverse tunneling (RFC 2344)





Mobile IP with reverse tunneling

- **Routers accept often only “topologically correct” addresses (firewall!)**
 - ❑ a packet from the MN encapsulated by the FA is now topologically correct
- **Multicast and TTL problems solved**
- **Reverse tunneling**
 - ❑ opt ded through the tunnel via the HA to a sender (longer routes)
- **The new standard is backwards compatible**
 - ❑ the extensions can be implemented easily





Mobile IP and IPv6

- **Mobile IP was developed for IPv4, but IPv6 simplifies the protocols**
 - ❑ security is integrated and not an add-on, authentication of registration is included
 - ❑ COA can be assigned via auto-configuration (DHCPv6 is one candidate), every oconfiguration
 - ❑ no need for a se rform router
- ❑ **MN HA not needed in this case (automatic path optimization)**
- ❑ **„soft“ hand-over, i.e. without packet loss, between two subnets is supported**
 - MN sends the new COA to its old router
 - the old router encapsulates all incoming packets for the MN and forwards them to the new COA
 - authentication is always granted





Problems with Mobile IP

➤ Security

- ❑ authentication with FA problematic, for the FA typically belongs to another organization
- ❑ no protocol for key management and key distribution has been standardized in the Internet
- ❑ patent and expor



spe

g)

➤ QoS

- ❑ many new reservations in case of RSVP
- ❑ tunneling makes it hard to give a flow of packets a special treatment needed for the QoS

➤ Security, firewalls, QoS etc. are topics of current research and discussions!





Security in Mobile IP

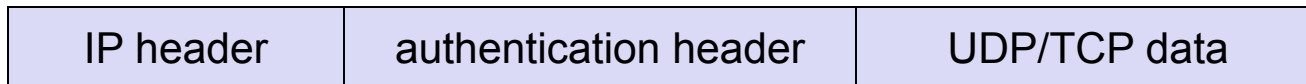
- **Security requirements (Security Architecture for the Internet Protocol, RFC 1825)**
 - ❑ **Integrity**
any changes to data between sender and receiver can be detected by the receiver
 - ❑ **Authentication**
 - ❑ **Confidentiality**
only sender and receiver can read the data
 - ❑ **Non-Repudiation**
sender cannot deny sending of data
 - ❑ **Traffic Analysis**
creation of traffic and user profiles should not be possible
 - ❑ **Replay Protection**
receivers can detect replay of messages





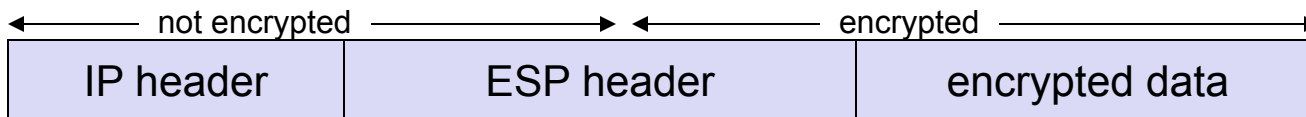
IP security architecture 1/2

- Two or more partners have to negotiate security mechanisms to setup a security association
 - ❑ typically, all partners choose the same parameters and mechanisms
- Two headers have been defined for securing IP packets:
 - ❑ Authentication-H



- ❑ Encapsulation Security Payload

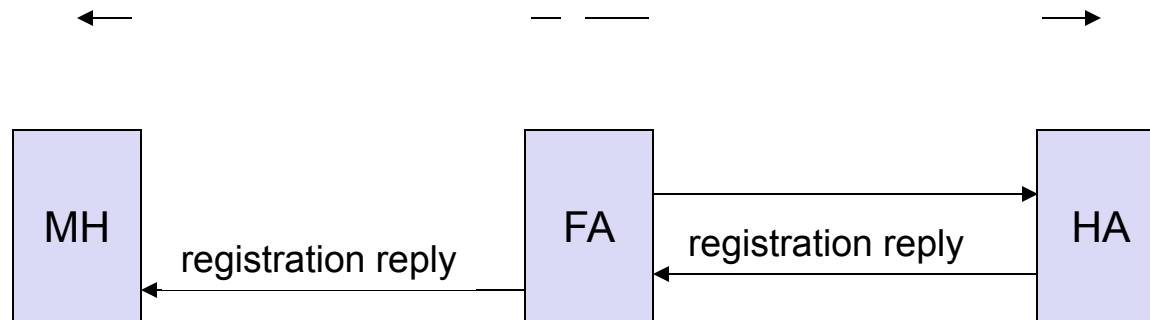
- protects confidentiality between communication partners





IP security architecture 2/2

- **Mobile Security Association for registrations**
 - ❑ parameters for the mobile host (MH), home agent (HA), and foreign agent (FA)
- **Extensions of the IP security architecture**
 - ❑ **extended authentication of registration**



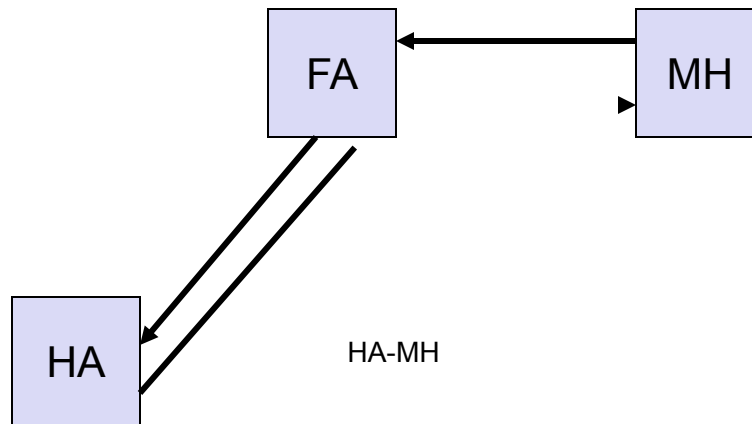
- ❑ **prevention of replays of registrations**
 - time stamps: 32 bit time stamps + 32 bit random number
 - responses: 32 bit random number (MH) + 32 bit random number (HA)





Key distribution

- Home agent distributes session keys



- foreign agent has a security association with the home agent
- mobile host registers a new binding at the home agent
- home agent answers with a new session key for foreign agent and mobile node





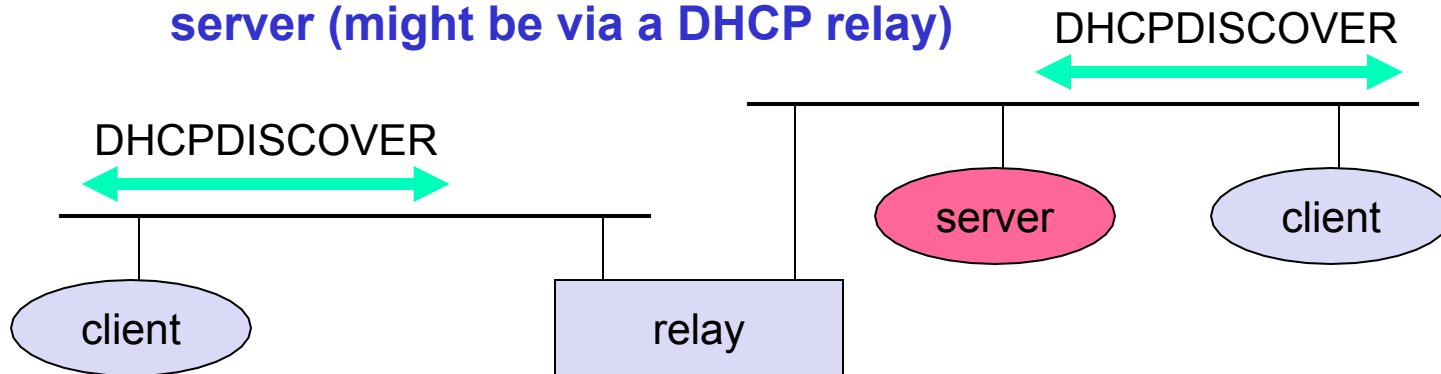
DHCP: Dynamic Host Configuration Protocol

➤ Application

- ❑ simplification of installation and maintenance of networked computers
- ❑ supplies systems with all necessary information, such as IP address, DNS server address, domain name, subnet mask, default router etc

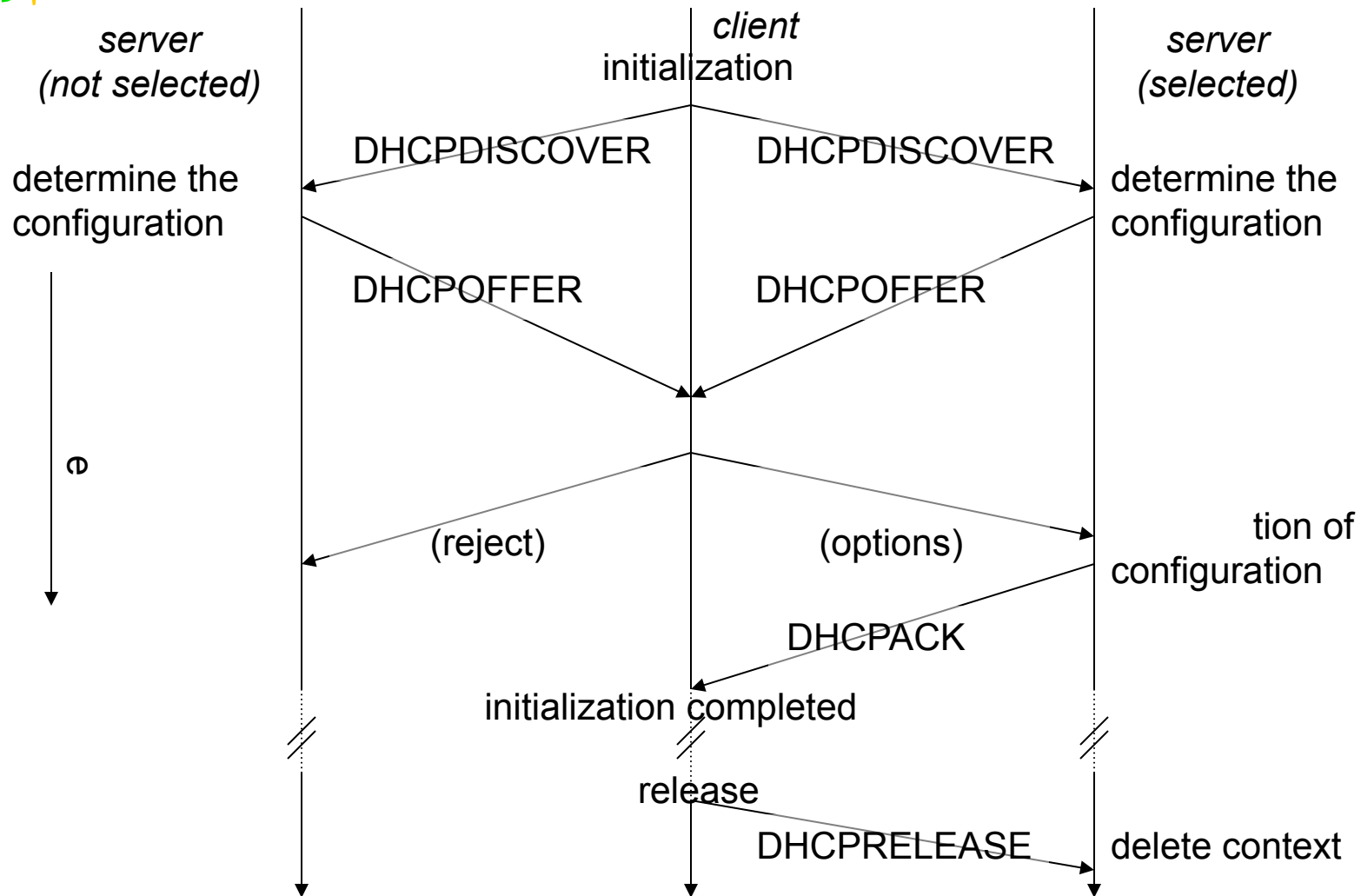
➤ Client/

- ❑ the client sends via a MAC broadcast a request to the DHCP server (might be via a DHCP relay)





DHCP - protocol mechanisms





DHCP characteristics

➤ Server

- ❑ several servers can be configured for DHCP, coordination not yet standardized (i.e., manual configuration)

➤ Renewal of configurations

- ❑ IP addresses have to be requested periodically, simplified protocol

- ❑ available
pro
DNS (domain name system) e
directory,

➤ Big security problems!

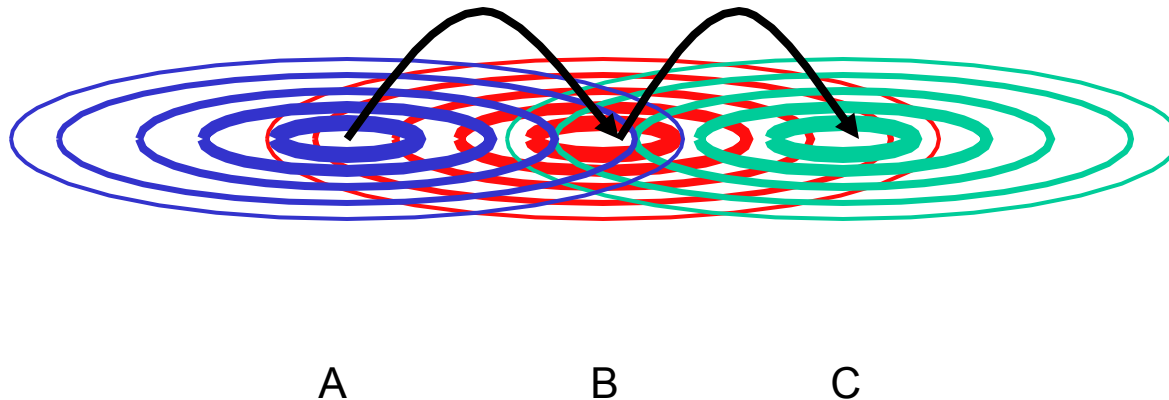
- ❑ no authentication of DHCP information specified





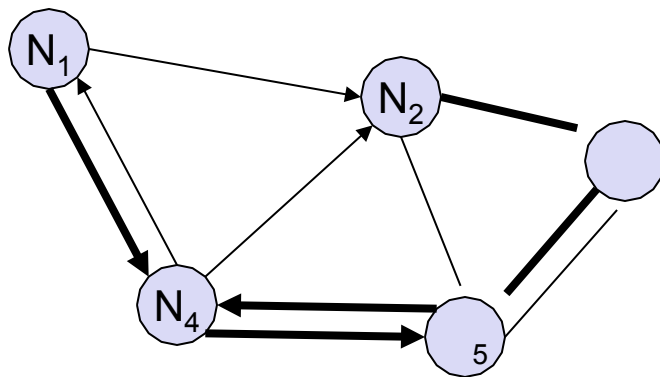
Ad hoc networks

- **Standard Mobile IP needs an infrastructure**
 - ❑ Home Agent/Foreign Agent in the fixed network
 - ❑ DNS, routing etc. are not designed for mobility
- **Sometimes there is no infrastructure!**
 - ❑ remote areas, ad hoc areas
 - ❑ no infrastructure
 - ❑ every node should be able to forward



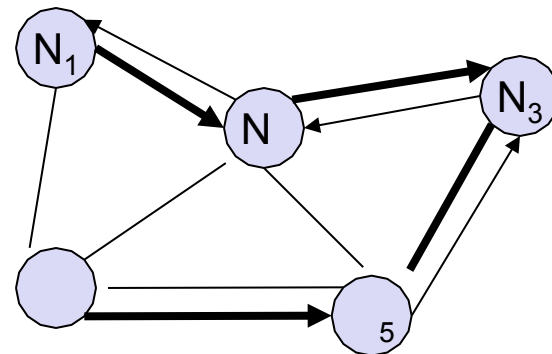


Routing examples for an ad-hoc network



time = t_1

→ good link
→ weak link



time = t_2





Traditional routing algorithms

➤ Distance Vector

- ❑ periodic exchange of messages with all physical neighbors that contain information about who can be reached at what distance
- ❑ selection of the shortest path if several paths available

➤ Link State

- ❑ rou

➤ Example

- ❑ **ARPA packet radio network (1973), DV-Routing**
 - every 7.5s exchange of routing tables including link quality
 - updating of tables also by reception of packets
 - routing problems solved with limited flooding





Problems of traditional routing algorithms

➤ Dynamics of the topology

- ❑ frequent changes of connections, connection quality, participants

➤ Limited performance of mobile systems

- ❑ periodic updates energy without

- ❑ limited bandwidth due to the

- ❑ links can be asymmetric, i.e., they can have a direction dependent transmission quality

➤ Problem

- ❑ protocols have been designed for fixed networks with infrequent changes and typically assume symmetric links





DSDV (Destination Sequenced Distance Vector)

- **Expansion of distance vector routing**
- **Sequence numbers for all routing updates**
 - ❑ assures in-order execution of all updates
 - ❑ avoids loops and inconsistencies
- **Decrease of updat**

tim





Dynamic source routing I

- Split routing into discovering a path and maintainig a path
- Discover a path
 - only if a path for sending packets to a certain destination is needed and no path is currently available
- Maintaining a path

- No periodic updates needed!





Dynamic source routing II

➤ Path discovery

- ❑ broadcast a packet with destination address and unique ID
- ❑ if a station receives a broadcast packet
 - if the station is the receiver (i.e., has the correct destination address) then return the packet to the sender (path was collected in the packet)
 - if the packet has a sender (identified via ID) then

- ❑ send a packet to the destination (with a list of addresses)

➤ Optimizations

- ❑ limit broadcasting if maximum diameter of the network is known
- ❑ caching of address lists (i.e. paths) with help of passing packets
 - stations can use the cached information for path discovery (own paths or paths for other hosts)





Dynamic Source Routing III

➤ Maintaining paths

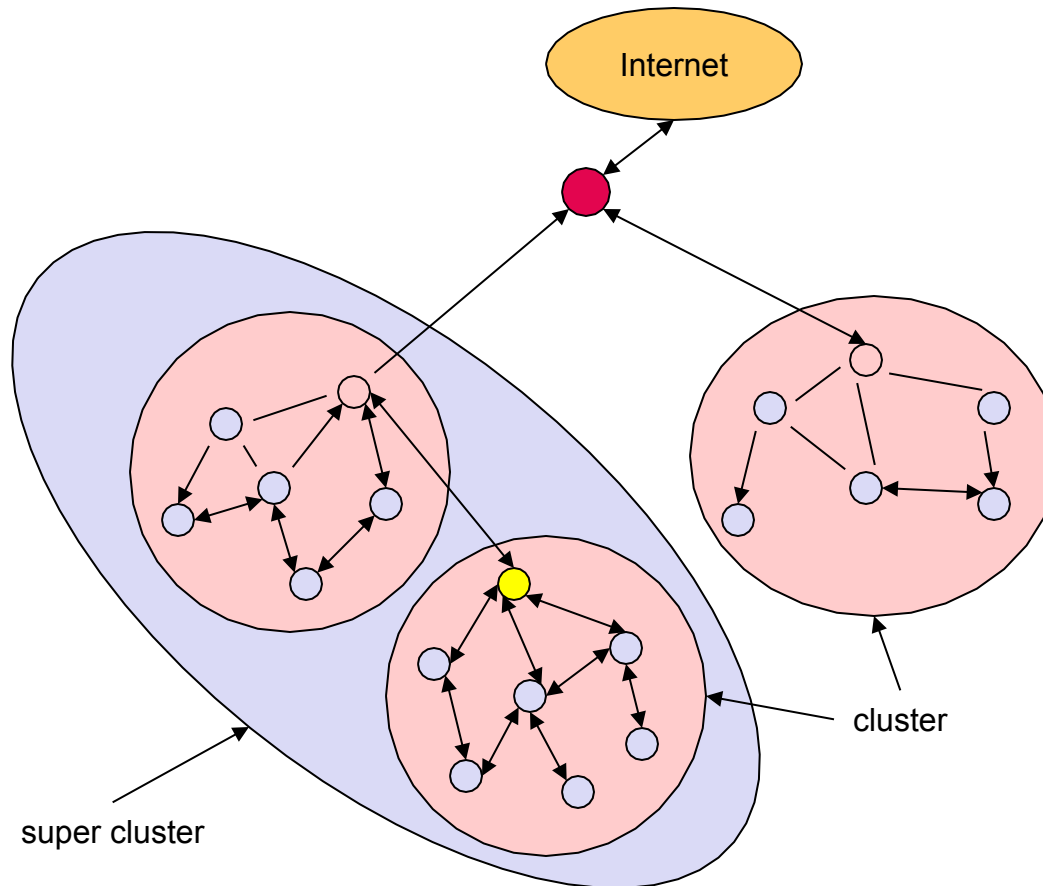
□ after sending a packet

- wait for a layer 2 acknowledgement (if applicable)
- listen into the medium to detect if other stations forward the packet (if possible)
- request an expli





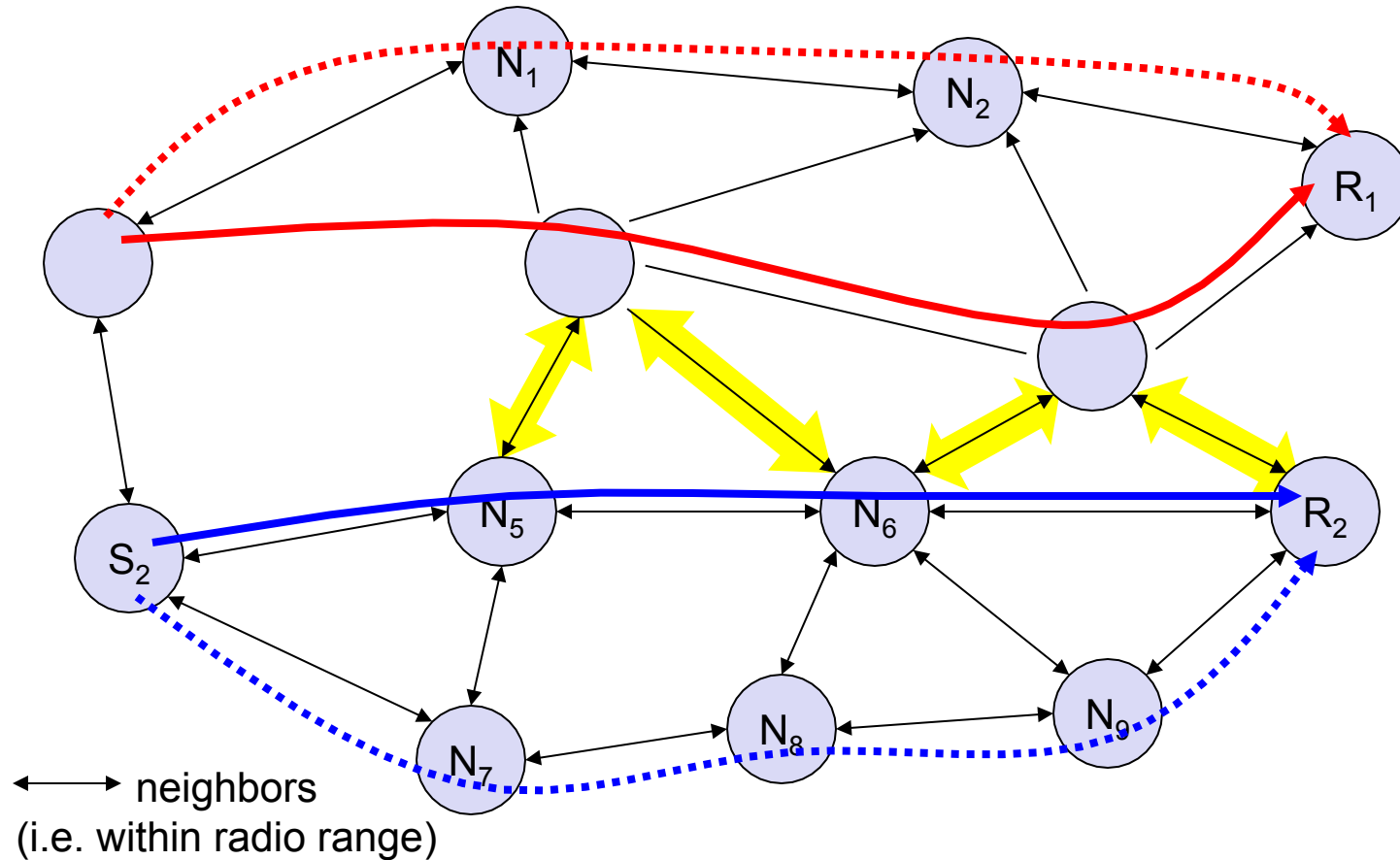
Clustering of ad-hoc networks





Interference-based routing

- Routing based on assumptions about interference between signals





Examples for interference based routing

- **Least Interference Routing (LIR)**
 - ❑ calculate the cost of a path based on the number of stations that can receive a transmission
- **Max-Min Residual Capacity Routing (MMRCR)**
 - ❑ calculate the cost of a path based on a probability function of successful trans nce
 - ❑ cal an ming
- **LIR is very simple to implement, only information from direct neighbors is necessary**

